

ERASTEEL TECHNICAL DATASHEET: ASP 2042

Application Family: Cold work | Standards: AISI M42 PM Equivalent, W.-Nr. 1.3247 PM

1. Grade Overview & Characteristics

ASP 2042 combines the high cobalt chemistry of M42 with powder metallurgy cleanliness, yielding significantly higher fatigue resistance and chipping resistance than ingot-cast M42 while maintaining peak hot hardness.

2. Nominal Chemical Composition (wt. % range)

Element	Carbon (C)	Chromium (Cr)	Molybdenum (Mo)	Tungsten (W)	Vanadium (V)	Cobalt (Co)	Other
Range %	1.05 - 1.11	3.6 - 4.0	9.2 - 9.6	1.4 - 1.8	1.1 - 1.3	7.8 - 8.2	Powder metallurgical version of M42

3. Mechanical Hardness & Property Profiles (Tempered state)

Metric	Value	Metric	Value (Rating 1-10)
Working Hardness Range	65.0 - 67.0 HRC	Machinability	6.0/10
Max Reachable Hardness	67.0 HRC	Wear Resistance	7.5/10
Annealed Delivery State	Max 280-350 HB	Toughness	6.5/10
Hot Hardness Retention	Secondary hardening	Hot Hardness	8.5/10
Heat Treatment Type	Vacuum / Salt bath	Grindability	6.5/10

4. Industrial Applications

Bi-metal saw edge wire, cold stamping punches for high-strength steel sheets, precision broaches, end mills.

5. Heat Treatment & Tempering Guidelines

Austenitizing at 1160-1190°C. Triple tempering at 560°C. Maximum working hardness 67 HRC.